The Impact of College Education on Old-Age Mortality: A Study of Marginal Treatment Effects

by

Evan Taylor
University of Michigan

CES 17-30 **March, 2017**

The research program of the Center for Economic Studies (CES) produces a wide range of economic analyses to improve the statistical programs of the U.S. Census Bureau. Many of these analyses take the form of CES research papers. The papers have not undergone the review accorded Census Bureau publications and no endorsement should be inferred. Any opinions and conclusions expressed herein are those of the author(s) and do not necessarily represent the views of the U.S. Census Bureau. All results have been reviewed to ensure that no confidential information is disclosed. Republication in whole or part must be cleared with the authors.

To obtain information about the series, see www.census.gov/ces or contact J. David Brown, Editor, Discussion Papers, U.S. Census Bureau, Center for Economic Studies 5K034A, 4600 Silver Hill Road, Washington, DC 20233, CES.Working.Papers@census.gov. To subscribe to the series, please click here.

Abstract

Using a newly constructed dataset that links 2000 U.S. Census long-form records to Social Security Administration data files, I evaluate the effect of college education on mortality. In an OLS regression, women and men who have at least some college education have 20% lower mortality rates than those with a high school degree or less. I proceed with an empirical design intended to illuminate the extent to which this relationship is causal, estimating marginal treatment effects (MTEs) using the proximity of the nearest college to individuals' birthplace as an instrument. Results indicate positive selection into college education (in terms of longevity) for both women and men. Selection drives almost all of the mortality gap for women. For men, longevity gains from college attendance are concentrated among individuals with unobserved variables that make them unlikely attend college. This suggests that men who would benefit most from receiving college education in terms of mortality reductions are those who are not attending.

[*]Any opinions and conclusions expressed herein are those of the author and do not necessarily represent the views of the U.S. Census Bureau. All results have been reviewed to ensure that no confidential information is disclosed. This is work in progress; suggestions are welcome. It is an essay in my dissertation, “Essays in Human and Social Capital" (Economics Department, University of Michigan). I would like thank my committee: Martha Bailey, John Dinardo, Andreas Hagemann, and Jeff Smith. I am grateful also for comments and help from Dan Black, Clint Carter, Seth Sanders, Bryan Stuart and Lowell Taylor, and participants in seminars at the University of Michigan. Finally, I wish to acknowledge the generous financial support of the NSF Graduate Research Fellowship Program.

1 Introduction

Well-educated people live longer and healthier lives than do their poorly-educated counterparts. This generalization holds across many countries, and has been true for more than a century [Cutler and Lleras-Muney, 2010]. The education-mortality gradient in the U.S. is dramatic; an analysis of 2000 mortality rates by Meara, Richards, and Cutler [2008] shows that the life expectancy for a 25-year-old with at least some college education was approximately seven years longer than for an individual with a high school degree or less. It appears that the education-mortality gradient is, if anything, increasing over time. However, while the relationship between education and mortality is well documented, we have only modest evidence about the extent to which it is causal.

My contribution is an analysis evaluating the impact of education on mortality in the U.S. Like Meara et al. [2008], I focus attention on mortality among those who have attended at least some college, in comparison to those with high school education or less. This is a highly relevant margin of schooling progression; among cohorts I study over 30% of women and 40% of men attained education above the high school level.[1] In the U.S., we are currently engaged in an active debate concerning policies that promote college education. Benefits in terms of improved health and increased longevity may represent a substantial portion of the total gains of college education, so an understanding of causal impacts is crucial for this debate.

My empirical approach is designed to take on two important empirical challenges that arise when analyzing causal effects of education on such outcomes as income, health, or mortality: First, individuals who receive higher education differ along many unobserved characteristics that may impact the outcomes; put differently, there is selection into college education. Second, the effect of education on outcomes may

[1] Among individuals now in prime working years, more than 60% have at least some college.

vary across individuals, and, in particular, may differ for those who have unobserved characteristics which make them more or less likely to obtain higher education. This type of heterogeneity in the treatment effect is called "essential heterogeneity" by Heckman, Urzua, and Vytlacil [2006b].

I use an instrumental variable approach that relies on distance to the nearest college or university (at age 17) from an individual's birth county. This instrument was used by Card [1995] to estimate the effect of education on earnings, and proximity to a nearby college has served as an instrument in many subsequent empirical analyses, including Cameron and Taber [2004] and Carneiro et al. [2011].

To calculate treatment effects which may be heterogeneous in both unobserved and observed variables, I employ estimators of marginal treatment effects (MTEs) developed by Heckman and Vytlacil [1999, 2005, 2007]. The estimation of MTEs provides improved conceptual clarity about impacts of education on mortality. As noted by Carneiro, Heckman, and Vytlacil [2011], understanding how treatment effects vary across individuals is crucial in evaluating the marginal benefits of any policy which may increase access to college education. For example, it may be that gains are highest among individuals who already would attend college, i.e., the average benefits realized by individuals induced to attend college by the expansion policy are lower than the average benefits of those who would have attended anyway.

I estimate the MTE in two different ways—an approach using local instrumental variables, which evaluates the derivative of the expected outcome with respect to the estimated propensity score, and the separate estimation approach discussed in Heckman and Vytlacil [2007], and implemented in the recent work of Brinch, Mogstad, and Wiswall [forthcoming, 2016]. The separate approach involves the use of stronger parametric assumptions, with a corresponding benefit in terms of increased precision on estimation. This proves valuable in my application.

My analysis uses a newly constructed dataset that links the 2000 U.S. Census to NUMIDENT records from the U.S. Social Security Administration.[2] These data include records for almost 5 million individuals born in the United States, 1911–1940. The data contain educational attainment (recorded in the 2000 Census) and place of birth, date of birth, and date of death for the deceased (from the NUMIDENT file).[3] The analysis of mortality focuses on the time span 2000-2010.

Consistent with the literature, I find a strong association between college education and mortality for both men and women. In an OLS specification, individuals with at least some college have a 10-year mortality rate 20% lower than those with a high school degree or less.

I implement a simple selection test proposed by Black, Joo, LaLonde, Smith, and Taylor [2016]. There is strong evidence of positive selection on longevity into college education for women. That is, women with a higher unobserved inclination towards education tend to have lower mortality whether they receive college education or not. For women, selection explains all of the differences in observed mortality rates between high and low education groups. For men, I only find evidence of positive selection on the *untreated* outcome. I do not find selection on the treated outcome, which suggests a "reverse Roy" pattern of selection for men.

In the MTE analysis, I find a heterogeneous treatment effects of college on mortality. For women, I estimate no treatment effect on old-age mortality. For men, MTE results show "essential heterogeneity" in treatment effects; individuals with unobserved characteristics that make them more likely to attend college have smaller treatment effects on mortality. I estimate an average treatment effect on mortality of zero for individuals who had at least some college, but find that college attendance would have reduced mortality by about 15% among individuals who did not attend

[2]NUMIDENT is an acronym for the Social Security Administration's "Numerical Identification System."

[3]I would like to acknowledge several colleagues with whom I am collaborating on the place-of-birth matching algorithm used in constructing data files: Martha Bailey, Bryan Stuart, and Reed Walker.

college. Taken at face value results suggest that for the cohorts of men I study, a policy that increased college attendance at the margin would have produced a substantial decline in mortality among affected men.

Studies such as Carneiro, Heckman, and Vytlacil [2011] and Moffitt [2008] suggest that the impact of college education on wages is concentrated among individuals whose unobservable characteristics make them most likely to attend college. Both papers find that individuals sort into college based on gains. These MTEs of education on income, in combination with my results, are consistent with Cutler, Deaton, and Lleras-Muney [2006] contention that the positive impact of education on longevity is likely not through any effect on adult income. I find that the individuals with little gains in terms of income receive large benefits in mortality.

My work contributes to an emerging literature that uses modern econometric methods to evaluate the effect of education on health and mortality. Much of the previous work has used changes in mandatory schooling laws, which exploit variation in the lower parts of the schooling distribution. The results of these studies have been mixed. Lleras-Muney [2005] finds large negative effects of increased education on mortality in the U.S. for whites born 1901–1925. Mazumder [2008] and Black, Hsu, and Taylor [2015a] find no evidence of causal effects using somewhat different data and/or approaches. Albouy and Lequien [2009] and Clark and Royer [2013] estimate the effect of education in France and the UK, respectively, and find no evidence of an effect. Using Swedish data, Meghir, Palme, and Simeonova [2012] find that negative effects on mortality induced by changes in compulsory schooling are concentrated among men from low SES backgrounds.

Also related to my study is recent empirical work from Buckles, Hagemann, Malamud, Morrill, and Wozniak [2016], who use variation in draft avoidance behavior during the Vietnam War and find that college education reduces mortality. In con-

trast to my work, Buckles et al. [2016] focus on mortality among those aged 28–65, and study men only (women were not subject to the military draft). Also, they did not estimate MTEs.[4]

The paper proceeds as follows: In the next section, I discuss the literature on the relationship between education and health. In section 3, I explain the definition and estimation of marginal treatment effects. In section 4, I develop a model of education and mortality which fits the MTE framework. The next section describes the data. In sections 6 and 7, I show results from the selection test and main results from the MTE analysis. The following section gives discussion of the results and possible mechanisms. The final section concludes.

2 Relationship between Education and Health

The positive correlation between education and health is well documented. Many factors contribute.

First, there are many reasons why even in the absence of a causal effect of education on health, we may see an association between the two. Adverse health circumstances experienced in infancy and early childhood can harm prospects for learning and reduce human capital accumulation,[5] and the same health deficits that impair learning early in life likely have an adverse impact on adult health. For instance, poor childhood nutrition may affect adult health [Fogel, 1997, 2004], as might adverse health events *in utero* [Barker, 1998]. Further, expectations about increased longevity could induce individuals to acquire higher levels of human capital [Ram and Schultz, 1979].

In addition, models of human capital investment—pioneered by Becker [1967], Mincer [1974], and Griliches [1977], and further developed by Card [2001], Heckman,

[4]Buckles and co-authors uses data at a cohort by birth-state level, and have two endogenous regressors (college and veteran status), making MTE estimation impossible.

[5]See Currie [2009, 2011] and Almond and Currie [2011] for comprehensive discussions of the evidence on this point.

Lochner, and Todd [2006a] and Heckman, Humphries, and Veramendi [2016], among others—lead us to understand that along unobservable dimensions, well-educated individuals differ from those with less education. There are two distinct forms of systematic selection into higher education at work. People may be *sorting on gains* (e.g., if those who have the most to gain are most likely to acquire schooling) and there may be *selection bias* unrelated to gains (e.g., if there is an individual fixed effect in the earnings function that is correlated with completed schooling). Even if education has no direct impact on health and longevity, we would observe an education-mortality gradient if components of income-related selection are correlated with health-related outcomes.

Finally, there are many theories as to why education may have a causal impact on health and mortality. The theory of human capital accumulation has been adapted to investments in health by Grossman [1972, 2000], Rosenzweig and Schultz [1982], Cutler and Lleras-Muney [2010] and Lochner [2011], among others.[6] Grossman [1972] argues that education can directly impact mortality and health by increasing the marginal productivity of health inputs, which he calls "productive efficiency." Cutler and Lleras-Muney [2010] suggest a number of channels through which education can improve health outcomes, and note that "better educated people are less likely to smoke, less likely to be obese, less likely to be heavy drinkers, more likely to drive safely and live in a safe house, and more likely to use preventative care."[7] Many studies provide additional evidence on this topic, including Conti, Heckman, and Urzua [2010], who identify heterogeneous beneficial causal effects of education on health behaviors, including smoking and exercise. Walque [2007] and Grimard and Parent [2007] find evidence that college education reduces smoking rates among men.

[6] Lochner [2011] offers a theoretical synthesis and overview of the literature.

[7] Cutler and Lleras-Muney [2010] are careful to note that these associations cannot be interpreted as causal in their framework. They do provide an informed judgment that "education seems to influence cognitive ability, and cognitive ability in turn leads to healthier behaviors. As best we can tell, the impact of cognitive ability is not so much what one knows but how one processes information. Everyone 'knows' that smoking is bad and seat belts are useful but the better educated may understand it better."

Education generally causes increases in earnings [Card, 1999]. If higher income allows individuals to afford better health care and nutrition, then education will have a positive effect on longevity. Cutler and Lleras-Muney [2010] suggest that about 20% of the education-health gradient can be explained by changes in economic resources. Caution is warranted when it comes to interpretation, however, because of the possibility of reverse causality; Smith [1999] argues that poor adult health causes decreases in income.

Clearly, the links between health and education are multi-faceted. One point of view, articulated by Cutler, Deaton, and Lleras-Muney [2006], suggests a central role for education: "It seems clear that much of the link between income and health is a result of the latter causing the former, rather than the reverse. There is most likely a direct positive effect of education on health." In an essay on the injustice of health inequality, Deaton [2013] provides a set of arguments for why it is important that we develop a conceptually clear empirically-grounded understanding of these relationships. For instance, if there is little impact of adult income on health, a transfer policy that increases income among the poor may have little impact of health inequalities.

3 Marginal Treatment Effects

3.1 Model

I first introduce the standard Marginal Treatment Effects framework used in Heckman and Vytlacil [1999]. The outcome variable is Y, which I define as a 10-year survival dummy. D is a binary indicator for treatment, which I define as 1 if the individual has at least some college education, and 0 if the individual is a high school graduate or less. I discuss the reasoning behind this definition in the next section. I have observable characteristics X and instruments Z. Let Y_1 denote the outcome when an

individual is treated and Y_0 denote the outcome when an individual is not treated. Crucially, when an individual is treated we only observe Y_1, and not the missing counter-factual Y_0, and vice-versa. Then, $Y = DY_1 + (1 - D)Y_0$. Suppose that the conditional mean functions for Y are as follows:

$$Y_{1i} = \mu_1(X_i) + U_{1i} \tag{1}$$

$$Y_{0i} = \mu_0(X_i) + U_{0i}, \tag{2}$$

where the vector X_i in empirical specification below include observable characteristics—birth state fixed effects, birth-county economic variables, cohort effects—that may affect survival in either treatment condition.

The indicator variable D_i reflects individual choice. The choice model is given by

$$D_i = \mathbb{1}(f(X_i, Z_i) > V_i); \tag{3}$$

the college attendance decision is influenced by both X_i and the instrument Z_i.

Let F_V be the CDF of V_i, $F_V(V_i) = U_{Di}$ and $F_V(f(X_i, Z_i)) = P(X_i, Z_i)$. We can rewrite (3) as

$$D_i = \mathbb{1}(P(X_i, Z_i) > U_{Di}), \tag{4}$$

where $P(X, Z) = E[D|X, Z]$ is the propensity score and U_D is distributed uniform over $(0, 1)$.[8] For Z to be a valid instrument, we need conditional independence: $(U_0, U_1, U_D) \perp\!\!\!\perp Z|X$. Vytlacil [2002] shows the equivalence of the latent index model shown above and the standard IV assumption of monotonicity [Angrist, Imbens, and Rubin, 1996].

In this set-up, it is surely possible that there will be a relationship between errors

[8]The use of the uniform distribution is a convenient normalization.

in the outcome equations (U_0, U_1) and the error in the choice equation, U_D. The term U_D represents all the unobservable determinants in the college attendance decision. These could include parents' wealth and educational attainment, latent "ability" or motivation, the individual's personal discount factor and childhood health. Going forward it is helpful to take note that in this formulation agents with low U_D are *more* inclined to attend college. Thus low U_D is likely associated with high parental SES, high ability, stronger childhood health, etc.

Figure 1 outlines three possible patterns of correlation between the error terms:

First, suppose that the treatment effect is homogeneous and individuals from advantaged backgrounds are generally healthier, i.e., have high values of both U_{0i} and U_{1i}, and are more likely to attend college. Thus, individuals with lower U_{Di} will tend to have $U_{1i} = U_{0i} > 0$. This pattern, shown in Panel A, is the familiar "ability bias" [Griliches, 1977]. In this instance, if we estimate the impact of college on survival in an OLS regression, we overstate the treatment effect.

Second, Panel B shows a case in which there is heterogeneity in gains from college attendance (in terms of increased longevity) and those individuals who have the most to gain attend college. In other words, there is selection on gains as in Roy [1951]. Here individuals with high values of $(U_{1i} - U_{0i})$ disproportionately have U_{Di} close to 0. Now it is possible that OLS understates the treatment effect of college among those most likely to attend college—those with U_D close to 0—while overestimating the potential value of college on those who do not attend.

Finally, Panel C illustrates an interesting alternative. Perhaps children from advantaged backgrounds build strong health capital and develop good health-related habits that persist into adulthood regardless of college education, *and* those same children are disproportionately likely to attend college (i.e., U_{0i} and U_{Di} are negatively correlated). Individuals that benefit the most from college are lower SES and

have high U_D. This has been called a "reverse Roy" selection pattern. In the example illustrated, OLS overstates the impact of college on those who attend college, but does not necessarily overstate the potential impact of college on those who are not currently attending. The MTE approach is designed to help us distinguish among possible patterns.

We work with the conditional expectation functions of the error terms,

$$c_1(x,p) = E[U_1|X = x, U_D = p] \tag{5}$$

$$c_0(x,p) = E[U_0|X = x, U_D = p]. \tag{6}$$

Then the marginal treatment effect is defined as

$$\Delta^{MTE}(x,p) = E[Y_1 - Y_0|X = x, U_D = p] \tag{7}$$

$$= \mu_1(x) - \mu_0(x) + c_1(x,p) - c_0(x,p). \tag{8}$$

The MTE is the expected difference in potential outcomes (i.e., the average treatment effect) for individuals who have observables $X = x$ and are indifferent between getting treatment and not when $P(X,Z) = p$.

3.2 Estimation through Local IV and Separate Approach

Define $K(x,p) \equiv E[Y|X = x, P(X,Z) = p]$. As shown by Heckman and Vytlacil [1999], the marginal treatment effect can be estimated by taking the derivative of the expectation of the outcome with respect to P, as follows:

$$\Delta^{MTE}(x,u_D) = \Delta^{LIV}(x,u_D) = \left.\frac{\partial K(x,p)}{\partial p}\right|_{p=u_D}. \tag{9}$$

This is the method of local instrumental variables.

Heckman and Vytlacil [2007] examine a second approach, which is based on estimating $E[Y|X,P,D=1]$ and $E[Y|X,P,D=0]$ separately rather than estimating $E[Y|X,P]$. In parametric estimation, the parametric assumptions for this approach are somewhat stronger. For example, if estimating an MTE which is linear in parameters, in the separate approach both $E[Y_1|X,U_D]$ and $E[Y_0|X,U_D]$ are assumed to be linear in X and U_D, whereas in Local IV only the difference, the MTE, is assumed to be linear. This can lead to large increases in precision of the estimates.[9]

Define K_1 and K_0 as such:

$$K_1(x,p) \equiv E[Y|X=x,P(X,Z)=p,D=1] = \frac{\int_0^p c_1(x,u_D)du_D}{p} + \mu_1(x) \tag{10}$$

$$K_0(x,p) \equiv E[Y|X=x,P(X,Z)=p,D=0] = \frac{\int_p^1 c_0(x,u_D)du_D}{1-p} + \mu_0(x). \tag{11}$$

Differentiating with respect to p and solving for c_1 and c_0 we get

$$c_1(x,p) = p\frac{\partial K_1(x,p)}{\partial p} + K_1(x,p) - \mu_1(x) \tag{12}$$

$$c_0(x,p) = -(1-p)\frac{\partial K_0(x,p)}{\partial p} + K_0(x,p) - \mu_0(x). \tag{13}$$

Plugging the above into (8), we get[10]

$$\Delta^{MTE}(x,p) = p\frac{\partial K_1(x,p)}{\partial p} + K_1(x,p) + (1-p)\frac{\partial K_0(x,p)}{\partial p} - K_0(x,p). \tag{14}$$

To generalize these two methods, define

$$\tilde{K}(x,p,d) \equiv E[Y|X=x,P(X,Z)=p,D=d] = dK_1(x,p) + (1-d)K_0(x,p). \tag{15}$$

[9] In unpublished work in progress, I discuss the relative merits of each estimation strategy and show solutions when the stronger parametric assumptions of the separate approach are incorrect [Taylor, 2016].

[10] Notice that both $\mu_j(X)$ cancel out. In general, these are not identifiable separately from K_j. See Heckman and Vytlacil [2007].

Then

$$\Delta^{MTE}(x, u_D) = \tilde{K}(x, u_D, 1) - \tilde{K}(x, u_D, 0) + \left.\frac{\partial \tilde{K}(x, p, 0)}{\partial p}\right|_{p=u_D} + u_D\left(\left.\frac{\partial \tilde{K}(x, p, 1)}{\partial p}\right|_{p=u_D} - \left.\frac{\partial \tilde{K}(x, p, 0)}{\partial p}\right|_{p=u_D} \right). \quad (16)$$

To estimate the MTE, we estimate the function $\tilde{K}(x, p, d)$, and substitute it into the equation above.

4 Model of Education and Mortality

I introduce and develop a simple model of educational attainment and mortality. The model builds on the canonical model of Becker [1967]. Individuals maximize expected utility $U(y, S)$, which is a function of earnings (y) and years of schooling (S). For my purposes, y can include the value of health and longevity, as in Lochner [2011]. Alternatively, it may be that individuals are unaware of benefits from education in terms of health improvements or increased longevity, or in any event that they do not take them into consideration when making human capital investments. For simplicity, let utility take a separable form, $U_i(y_i, S_i) = f(y_i) - \phi_i(S_i)$, and then let $y_i = g_i(S_i)$, as in Card [1994]. $\phi_i(S_i)$ represents the total costs of schooling, including monetary and non-monetary costs. Finally, let $m_i(S_i)$ represent the individuals expected mortality rate as a function of their schooling.

In my research design, the key assumption is that the instrument, college proximity, alters the cost curve, $\phi_i(\cdot)$, but does not have an impact on the mortality function $m_i(\cdot)$ nor the function that generates returns from schooling $g_i(\cdot)$.

Note that for the instrument to be valid, it can only shift individuals *across* the treated or untreated states and cannot shift people to different educational attain-

ments that are *within* either the treated or untreated state. This follows from the MTE assumption that $(U_1, U_0) \perp\!\!\!\perp Z|X$.[11] If we rewrite the schooling cost curve, $\phi_i(S_i) = \phi_i'(S_i) + \lambda_i \mathbb{1}(S_i > 12)$, then the instrument can only affect λ_i, but not $\phi_i'(\cdot)$. The assumption is that the distance to the nearest college will only affect the fixed cost of first attending college, while having little impact on the marginal cost of each year of school thereafter. Similarly, the instrument must also not affect the marginal cost curve for $S < 12$.

This is plausible if most of the cost reduction coming from proximity is a one-time "transition cost" (e.g., migration cost), which includes all monetary and non-monetary costs. I will return to concerns about biases that are introduced if college proximity also affects marginal costs beyond the first year of college.

This model guides the choice of the definition of the treated state as "some college" and the untreated state as "high school or less." If the cutoff for treatment was defined as at least a college degree and some college was part of the untreated state, then the instrument could not affect the initial cost of going to college and only the margin between attending and graduating. This latter set of assumptions is less credible.

For each individual, there is an optimal level of schooling that is less than or equal to 12-years (high school or less), and an optimal level of schooling that is greater than 12 years (at least some college). These can be written as $S_{0i} = \arg\max_{S \leq 12} U_i(y, S)$ and $S_{1i} = \arg\max_{S > 12} U_i(y, S)$. Then $Y_{0i} = m_i(S_{0i})$ and $Y_{1i} = m_i(S_{1i})$. The individual attends college if $U_i(y_i, S_{1i}) > U_i(y_i, S_{0i})$. Figure 2 shows an example of the cost curve, $\phi(S)$ and benefit curve $f(g(S))$. The instrument can make parallel shifts of the cost curve for $S > 12$, which do not affect the optimal conditional choices S_0 and S_1. Thus, the treatment effect does not simply encompass the marginal gain between 12 and 13 years of school. The treatment effect evaluates the gain from college

[11] Alternatively, it could be the case that the mortality function $m_i(S_i)$ is flat except for a discontinuity at $S_i = 12$. Given the results from the health literature which show constant gains in health throughout the education distribution, see Cutler and Lleras-Muney [2010], this makes the latter improbable.

attendance, relative to non-attendance, for an individual who makes the preferred schooling decision within each treatment state.

5 Data

5.1 Census and NUMIDENT

As discussed in the introduction, many papers measuring the effect of education on mortality in the U.S., such as Lleras-Muney [2005], Black et al. [2015a], Buckles et al. [2016], have exploited variation at a state-by-cohort level, and measured state-by-cohort mortality rates using census and vital statistics data. While these strategies can be used to estimate an IV, they are unsuitable for the evaluation of essential heterogeneity in treatment effects. To estimate MTE, we require a large sample of individual level data which contains both information on educational attainment and mortality, and we need an instrument which can be measured at the individual level at an early age.

The primary data in the analysis comes from two sources, the 2000 U.S. Census and the NUMIDENT from the Social Security Administration. The two sources have been linked using the Protected Identification Key (PIK). Every individual in the NUMIDENT file is given a unique PIK, which is then matched to the 2000 Census based on personal information.

The 2000 U.S. Census contains information on completed years of education, race, gender and age. The NUMIDENT has basic demographic information, as well as the individual's date of birth and date of death if they are deceased. The NUMIDENT also contains a 12-character place of birth field and a 2-character state of birth field. This is used to match the individual to a physical location. The algorithm used to perform this match is an augmented version of the algorithm my co-authors and I developed in Black, Sanders, Taylor, and Taylor [2015b]. We take place names from

the U.S. Geological Service's Geographic Names Information System (GNIS). The GNIS is a list of all place names in the United States, including both current and historic. From this we match individuals to a county of birth.[12] A relatively small percentage of individuals cannot be matched to place of birth, either because the place of birth field is missing, because we cannot determine their place of birth from the 12-character string, or because do not have PIKs and cannot be linked to the NUMIDENT. These people are not included in the analysis.

My sample contains almost 5 million white individuals born in the continental United States from 1911–1940, who are in the 2000 U.S. long-form Census and can be linked to a birthplace through NUMIDENT.[13] I do not include individuals that have imputed education, whose age in the Census was five or more years different than their age in the NUMIDENT, and whose race or gender was not the same in the Census and the NUMIDENT.[14]

Of the 2.7 million women in the data set, 34% attended at least some college. Men in the data attended college at higher rates; 43% of the 2.2 million men had at least some college. For the dependent variable in my analysis, I use a 10-year survival indicator from 2000–2010. The indicator is equal to zero if the individual is deceased in 2010 or earlier, and one otherwise. In the sample, the average 10-year mortality rate was 32% for women and 39% for men.

5.2 College Data

The college data come from the Integrated Postsecondary Education Data System (IPEDS), published by the National Center for Education Statistics (NCES). The IPEDS data extend back to the year 1980, so the data contain all 2-year and 4-year

[12]For individuals who were born in a county that has changed borders, the algorithm will match to the current county of the individual's birthplace.

[13]A separate analysis for blacks showed the instruments were weak, even when restricting the distance measure to historically black college and universities. More work is needed to estimate the effect of college education on blacks.

[14]When there is a discrepancy between birth year in the Census and NUMIDENT, I use the NUMIDENT birth year.

colleges that were open in the United States in 1980, but not colleges that closed prior to 1980.[15] IPEDS contains information on the location as well as the year that the college opened.

5.3 Instrument

Using these data, I construct the instrument—proximity from the individual's county of birth to the nearest college, including both two and four-year colleges. From the NUMIDENT data it is possible for most individuals to construct a city of birth rather than county of birth. However, for all individuals, I use distance from the individual's birth county, even if it is possible to determine the city of birth. The county of birth measure is less likely to violate the exclusion restriction, given that all economic covariates are at a county level. This approach also allows for geographic sorting within a county. To approximate the average distance faced by a resident of a county, I calculate the distance from the centroid of the county to the nearest college. The measure includes all colleges that are open when the individual turns 17.

There are several potential threats to exogeneity of the instrument. First, parents who have a strong preference towards sending their children to higher education may choose to live in counties nearer to colleges. If the children of these parents are also of higher "ability" or are different in some other unobserved way that is correlated with later life mortality, then the exclusion restriction would be violated. In this situation, all estimated treatment effects would be biased upwards. If these parents instead choose to live in places that are observably wealthier without thought to how close the nearest college is, then the instrument remains valid.

Secondly, places that have colleges may be healthier for reasons beyond the effect of increased education for the individuals that are born there. If, controlling for other

[15]This may affect the power of the instrument, but not the exclusion restriction as long as locations that had a closed college are no different than locations that never had a college along unobservable characteristics that affect mortality.

wealth measures, counties near colleges are healthier—perhaps because they have better access to medical care or because they have better primary and secondary public education—then the exclusion restriction would be violated. Again, in this case, estimated treatment effects would be biased upwards.

A third possible violation is that the presence of college can shift individuals not only across the no college versus some college threshold, but to different educational attainment categories within these groups. For example, consider an individual who was born far away from a college and dropped out of high school. It would be a violation of the exclusion restriction for that individual to get a high school degree in the state of the world where she is born close to the college. As shown in the model, the presence of the college can only affect the marginal cost of moving from the 12th year of school to the 13th year of school. In this case, the instrument will only shift individuals *across* the some college threshold. Note that this is a stricter set of assumptions than needed to use the instrument on a continuous measure of education. This assumption is necessary because of the use of the binary measure of education, which is needed to estimate a marginal treatment effect.

I control for several observed characteristics of the individual's birth county, to eliminate the potential effects of colleges being located in wealthier areas. These are an array of county-level economic characteristics from near the birth years and present day. From the Bureau of Economic Analysis, I use the average per capita income from 1969-2010 to control for the modern day wealth of a county. These data only go back to 1969, and the Census data on prosperity measures vary by decade. I include the median housing value in 1930, the percentage of households which owned a radio in 1940, and the median household income, by decile, in 1950.[16] While these old measures are likely somewhat noisy measures of wealth, they are generally strongly correlated with each other, as shown in Table 1.

[16]Where data is missing, I impute zeros and include an indicator variable for missing data in all specifications.

6 Selection Test

Black, Joo, LaLonde, Smith, and Taylor [2016] provide a simple selection test with a binary dependent variable and an instrument. The test is two separate OLS regressions for treated and untreated individuals—regressing the outcome Y on both X and Z:

$$Y_{1i} = \alpha_1 X_i + \beta_1 Z_i + \epsilon_{1i}, \quad \text{for } D_i = 1, \quad \text{and} \tag{17}$$

$$Y_{0i} = \alpha_0 X_i + \beta_0 Z_i + \epsilon_{0i}, \quad \text{for } D_i = 0. \tag{18}$$

$\beta_1 \neq 0$ implies selection on unobserved variables on the treated outcome. That is, the unobserved variables that enter the choice equation are correlated with the unobserved variables that enter the treated outcome equation. Similarly, $\beta_0 \neq 0$ implies selection on the untreated outcome.

If the instrument is valid, in the absence of selection $E[Y_1|X, Z]$ and $E[Y_0|X, Z]$ are determined by X only. With positive selection on Y_1, we expect that as we decrease the distance to college we increase the marginal propensity score and move further into the U_D distribution. This in turn decreases the expected value of the outcome, as in Figure 3. Thus, in this case $\beta_1 > 0$. Similar intuition shows that with positive selection on the untreated outcome, Y_0, we have $\beta_0 > 0$.

These tests also provide a sanity check on the validity of the exclusion restriction. Given the evidence discussed in the prior literature, our prior on the nature of selection into education suggest that it should be weakly positive on both the treated and untreated outcomes. Therefore, $\beta_1 < 0$ and $\beta_0 < 0$ imply that either selection into education is negative or that the exclusion restriction is violated such that counties further from colleges have lower survival rates. This would constitute evidence that the exclusion restriction is violated, as negative selection into college on health goes

strongly against our priors.

6.1 Results for the Selection Test

Estimates for the selection tests for women and men appear in Table 2. The results for women show evidence of positive selection on both the treated and untreated outcomes. For men, there is strong evidence of positive selection on the untreated outcome, but no evidence of selection on the treated outcome. Any bias coming from the instrument, in which individuals born in closer proximity to colleges are more likely to live longer, would bias these coefficients downward, making it more difficult to find positive selection. These tests provide credible evidence of positive selection into education in terms of mortality.

Results from the selection tests foreshadow the MTE analysis. Consider results for men. Among those who attend college, expected longevity Y_1 (conditional on X) is approximately the same for those who are born close to a college and those born far from a college ($\beta_1 \approx 0$). Among who do not attend, expected longevity Y_0 is lower for men born close to a college than those born far away ($\beta_0 > 0$). These suggest a pattern of selection as in Panel C of Figure 1—"reverse Roy" sorting. The MTE curve for survival is upward-sloping in U_D. For women, the similarity in the coefficients from the selection regressions may imply a flatter MTE curve, and that selection likely explains a large portion of the observed differences in mortality between the treated and untreated.

7 Results

7.1 OLS Regression

To explore basic relationships, I begin with OLS regressions of college education on 10-year survival, using variants of the following specification:

$$Y_i = \alpha + \beta X_{c(i)} + \lambda_{b(i)} + \zeta_{s(i)} + \gamma D_i + \epsilon_i, \tag{19}$$

where $Y_i = 1$ if an individual survives past year 2010, and $Y_i = 0$ if the individual is deceased in year 2010 or earlier. $X_{c(i)}$ are birth county covariates, $\lambda_{b(i)}$ are cohort fixed effects, $\zeta_{s(i)}$ are birth state fixed effects, and $D_i = 1$ if the individual has at least some college.

Results are presented in Tables 3 and 4. Women with at least some college have a 10-year survival rate 6 percentage point higher than women with high school or less. This is roughly a 20% reduction in the 10-year mortality rate. The results for men are similar. Men who attended college have a 21% lower 10-year mortality rate than men who did not.[17]

Birth-county covariates and even birth-county fixed effects do not dramatically change the result after controlling for state level fixed effects. The coefficients on college for both men and women are slightly larger when including birth-county fixed effects than when just including birth-state fixed effects. This indicates that, conditional on education and birth state, individuals have a *higher* survival rate if they were born in a county with *lower* average college attendance. This is consistent with a story of positive selection. With positive selection, as the college attendance rate

[17]For a comparison, Lleras-Muney [2005] found in OLS that each additional year of schooling reduces 10-year mortality rate by 0.036, using 1901-1925 birth cohorts in the U.S. with mortality measured from 1960-1980. The difference in education between the high and low groups was four years, which suggest roughly 0.02 reduction in 10-year mortality per extra year of schooling. Lleras-Muney [2005] use state-cohort level data rather than individual level data, which makes a direct comparison difficult. However, it is consolatory that the estimates are not substantially different.

rises within a county, it digs further into the "ability" distribution, reducing the average survival rates of both treated and untreated groups. Without selection, this would indicate that counties with higher education rates, conditional on the birth state, were less healthy places. This seems implausible.

The estimates with birth-state fixed effects and birth-county fixed effects are fairly similar, which is somewhat reassuring for the validity of the instrument. If estimates from a birth-county fixed effects model were substantially lower that with birth-state fixed effects, it would indicate that individuals born in counties with high educational attainment had much better health outcomes, even conditional on their own education. This would be damaging evidence for the validity of the instrument.

7.2 IV Regression

For comparison to OLS and MTE results, I present instrumental variable regressions. For men and women, I estimate first stages using linear, quadratic and cubic polynomials in the distance to nearest college. First stage results are presented in Tables 5 and 6. F-stats from the different first stage models are between 22 and 37, well above the rule-of-thumb cutoffs for weak instruments [Stock, Wright, and Yogo, 2002]. For women, an increase in distance of 100 miles reduces college attendance by three percentage points, which amounts to an effect of approximately 9%. The instrument has a larger effect on men. A 100 mile increase in distance reduces college rates by five percentage points, approximately 12%.

IV results are presented in Table 7. In none of the specifications for men or women are the point estimates on college education significantly different than zero. If we made the (likely incorrect) assumption of a homogeneous treatment effect, these results would not provide clear evidence on the magnitude or sign of the effect. If treatment effects are heterogeneous, then IV estimates are a weighted average of

marginal treatment effects.[18]

7.3 MTE

I start with a simplifying assumption common in the estimation of MTEs that $(U_0, U_1, U_D) \perp\!\!\!\perp (X, Z)$, which implies that the MTE is additively separable in U_D and X.[19] I allow the treatment effect to vary on unobserved and observed variables, but restrict to no interactions between the observed and unobserved variables in the treatment effect. Here X includes birth-state and cohort fixed effects, as well as county level economic covariates.

To estimate the propensity score, I use a logit model. The model includes interactions between a cubic polynomial in the instrument and the birth state. I also include interactions between all covariates and decade of birth, to allow for the effect of all variables to vary over time. This specification includes a large number of parameters so we may be worried about weak instruments. However, results are robust to the first-stage model specification. Less flexible models raise concerns about misspecification of the first stage, which would bias our estimates of the MTE.

Densities of the propensity score estimates from this model are shown for women and men in Figures 4 and 5, respectively. Display of the histogram has been trimmed at 5% on both tails, per U.S. Census Bureau guidelines. The trimmed observations are still included in all analysis.

As discussed above, MTE can be estimated in two different ways: local instrumental variables and the separate approach, which estimates the second stage separately for treated and untreated individuals. When estimating a linear MTE, the parametric assumptions behind the separate approach are slightly stronger than for local IV.[20] In

[18] Heckman and Vytlacil [2005] provide formulas for weights. They note that weights are not always guaranteed to be non-negative.

[19] This assumption has been used in Carneiro et al. [2011] and Moffitt [2008] among others.

[20] Brinch et al. [forthcoming, 2016] show how these assumptions can be used to estimate a MTE in cases where local IV cannot, such as with a binary instrument.

the separate approach, both $E[Y_1|X, U_D]$ and $E[Y_0|X, U_D]$ are assumed to be linear in X and U_D, whereas in local IV only the difference, the MTE, is assumed to be linear. In this case, the slightly stronger assumption of the separate approach leads to huge efficiency gains, whether allowing for heterogeneity in the treatment in X or not. The separate approach also allows estimation of both conditional expectation curves, $E[Y_1|X, U_D]$ and $E[Y_0|X, U_D]$, whereas local IV only estimates the difference. Thus, the separate approach can inform us about the nature of heterogeneity and selection.

I estimate the marginal treatment effect model as follows:

$$\Delta^{MTE}(X, U_D) = \alpha + \beta U_D + \gamma X. \tag{20}$$

Results for the slope coefficient on U_D when using the separate estimation approach appear in Table 8. For women and men, the point estimates are positive. This suggests that individuals with unobserved characteristics that make them more likely to attend college had lower treatment effects on survival (recall the low U_D is associated with higher unobserved proclivity towards college). For men, results using the preferred first stage specification, flexible logit, are significant at 10% level. Results are similar when using a less flexible logit or linear probability model in the first-stage. In these specifications I include a cubic in the instrument, but do not include interactions of the instrument with birth-state. These specifications still include full interactions with decade of birth in the first-stage. All subsequent analyses use only the preferred flexible logit first-stage specification.

Table 9 shows estimates using local IV and the separate approach with and without allowing for heterogeneity in the treatment effect in X. Column (1) shows a constant effect model. Columns (2) and (3) show results using local IV. Standard errors are five to nine times larger using local IV than the separate approach. The slightly

stronger parametric assumptions under the separate approach induce huge efficiency gains. The standard errors in Column (3), which does not restrict $\gamma = 0$, are too large for meaningful interpretation of the parameters.

Columns (3) and (5) allow the treatment effect to vary on observed characteristics of the individuals birth county and state.[21] As in Moffitt [2008], allowing for heterogeneity on observed and unobserved variables results in much larger standard errors. For women, the sign of β flips when making the restriction that $\gamma = 0$. As it seems improbable that the treatment effect could vary on unobserved variables but not observed variables, estimates in columns (2) and (4) are given for completeness only. However, for men, the restricted and unrestricted models are almost identical under the separate approach.

7.4 Average Treatment Effects for Treated and Untreated

The marginal treatment effect is only identified for U_D over the support of the propensity score, $P(X, Z)$. However, using linear extrapolation we can calculate average treatment effects on the treated and untreated. The perils of such extrapolation are well known; results should be interpreted with caution. I calculate these effects because it adds substantial clarity to the patterns of selection and treatment heterogeneity found in the MTE analysis, and to give estimates of the average treatment effects implied by the results.

Results are presented in Table 10. The *treated* group is comprised of individuals with at least some college, and the *untreated* have a high school degree or less. Birth cohorts have been balanced across treated and untreated groups, so as to eliminate any effect rising educational attainment over time.

Results for women show average treatment effects are close to zero for both treated and untreated groups. Differences in survival rates between the treated and untreated

[21] Note that caution should be taken in interpretation of α in these models.

groups shows the strong patterns of positive selection into college education. My findings are consistent with an interpretation that selection explains essentially all of the observed difference in mortality rates between college-educated women and women with no college. This can be seen by comparing the differences in conditional outcomes between the groups (0.074 when untreated and 0.050 when treated) to the OLS estimate (0.061).

For men, we see evidence of positive selection only on the *untreated* outcome, mirroring the results found in the Black et al. selection test. I find evidence of a beneficial treatment effect for the individuals who did not attend college. The point estimate for the average treatment on the untreated is close to OLS result for men, suggesting a rather large effect.

Table 11 shows the same analysis, but with all observed covariates balanced across the untreated and treated individuals. Results are largely unchanged, showing that the selection and heterogeneity patterns found are almost completely driven by unobserved rather than observed characteristics. How this analysis would change if we could observe more individual-level characteristics which are observed in other data sets, such as parental education or some measure of ability, is an open question.

8 Mechanisms and Discussion

In summary, my study of marginal treatment effects imply a large positive effect for men who did not attend college, and no effect for those who did. I find no evidence of a causal impact of college education on the mortality of women. In this section, I explore implications of these findings and possible mechanisms consistent with the empirical results.

8.1 Income

While I cannot directly separate the effects of education and income, the MTE analysis provides suggestive evidence on the causal impact of income on mortality.

Using data from the NLSY, Carneiro, Heckman, and Vytlacil [2011] find the marginal treatment effect of college education on income is positive and decreasing in U_D. That is, individuals with unobservables that make them more likely to attend college are the ones who benefit the most in terms of income. With parametric models, Carneiro et al. estimate that the average treatment effect on the treated of a year of college is a 14% increase in earnings, and the average treatment on the untreated is zero. This is consistent with the findings of Moffitt [2008], who estimates the MTE of higher education on income using data from the UK.[22] In combination, these results and my findings imply that men who experience the largest income gains from college have the lowest benefits in terms of reduced mortality. The individuals who would have received near-zero gains in income from attending college had large longevity benefits. This is consistent with arguments of Cutler et al. [2006] and Deaton [2013], who believe that the impact of income on mortality is small.

8.2 Behavioral and Environmental

As noted earlier, there are a number of papers that discuss the potential pathways by which education might reduce mortality, and there is a literature that provides evidence about the causal impact of education on health behaviors and outcomes, such as smoking, drinking, and BMI. However, we know little about heterogeneity in the effect of education on these behaviors. It is possible that this heterogeneity is driving the results found here. For example, if individuals from high socio-economic back-

[22]For my analysis I do not have some of the variables observed in other studies, e.g., parental education. Thus, there is not a direct one-to-one correlation between the U_D in my work and the error term in those studies. However, they should be strongly correlated given that treatment status cannot change for any given observed variables. These papers also study later cohorts than the ones in my analysis.

grounds (which are associated with low U_D) are already predisposed not to smoke,[23] then it is plausible that the treatment effect of education on smoking is smaller for individuals with lower U_D. Further research is needed to know if the behavioral effects of education on health-related decisions and activities are heterogeneous across observed and unobserved variables.

For the cohorts I study, it is possible that education has an effect on the types of jobs individuals worked, which in turn had health impacts. There may have been important gender differences in any such relationships as well. Table 12 shows counts by gender and education from the 1960 Census for three different occupational categories that have may have been especially high risk in terms of health.[24] While these are a select group of occupations, the differences by gender and education are striking. Men from these cohorts without college education were much more likely to work in potentially dangerous factory or mining jobs. In all probability, men from advantaged socio-economic backgrounds, associated with low U_D, were less likely to work in the dangerous jobs regardless of their education. Men from poorer backgrounds, on the other hand, may have only been able to avoid working the most dangerous jobs if they acquired a college education. Women were less likely to be working in low-skill mining or smelting jobs irrespective of their education. Again, further research is needed to understand the causal effects of education on work-related health conditions.

8.3 Migration

There has been little attention to the role of lifetime migration in the relationship between education and mortality. Johnson and Taylor [2016] and Black, Sanders, Taylor, and Taylor [2015b] show that migration had a negative causal effect on older-

[23] Soteriades and DiFranza [2003] found that parental education and income were strongly negatively correlated with adolescent smoking rates.

[24] The differences in college attainment rates between this table and the 2000 Census arise from the differences in cohort distribution caused by mortality prior to 2000.

age mortality for individuals born in the early 20th century U.S. These papers also show that well educated individuals were more likely to migrate.[25]

With my data, I can provide some additional evidence about how education might impact migration. Using the same method to estimate MTEs of higher education on survival, I examine MTEs of college education on lifetime migration. I define lifetime mobility as living outside one's birth state as of 2000.

OLS results, shown in Table 13, indicate that higher educated people have higher lifetime migration rates. Conditional on education, individuals that migrate have lower survival rates. The estimated association between survival and migration is small relative to the association with college education, but it is non-trivial.

I estimate MTEs of college education on migration. Estimates of β are -0.531 (s.e. 0.096) for men and -0.409 (s.e. 0.094) for women. I estimate average effects of 0.355 (s.e. 0.079) for men and 0.262 (s.e. 0.090) for women. College education induces higher lifetime mobility,[26] and individuals with unobservables that make them most likely to get college education have larger average treatment effects of education on migration. Individuals with high U_D have only a small treatment effect of education on migration.

These results are consistent with the possibility that some of the heterogeneity in the impact of education on mortality could be the consequence of the auxiliary effect of education on migration. Individuals with low U_D have the largest effect of education on migration, which could mute any positive direct effect of education on mortality. However, this reasoning is somewhat speculative, because we cannot directly estimate the effect of migration on mortality in this framework. Johnson and Taylor [2016] suggests that the effect of migration on mortality is not homogeneous

[25] In ongoing work, Aaronson, Mazumder, Sanders, and Taylor [2016] find that the mortality benefits of Rosenwald Schools for African-Americans in the South were positive but counter-balanced by the negative effect of education on migration and migration on health.

[26] This result is consistent with Malamud and Wozniak [2012].

across the population. Furthermore, MTEs on migration do not differ much between men and women, so they do not explain gender differences in the effect of education on mortality.

8.4 Exclusion Restriction and Bias

As discussed above, most of the potential violations of the exclusion restriction point to an upward bias in the treatment effects. Thus results might best be interpreted as upper bounds on the true effects. However, all the average treatment effects estimated are near zero except for the treatment effect on men with high school education or less. If all estimates exhibit meaningful upward bias, this implies that for most groups the true effect of education is to *increase* mortality. This strongly flies against our priors on models of education and health. It also seems unlikely that the instrument would cause positive bias only in the estimated effect on low-educated men, while leading to unbiased estimated effects for the other groups.

9 Conclusion

In this paper I estimate the marginal treatment effects of college education on mortality. OLS estimates show a strong association between college attendance and reduced mortality. Positive selection into education, for both men and women, accounts for part of that association. I find no evidence of a causal effect of college education on mortality for women; all of the observed differences in mortality rates between education groups can be explained by selection. For men, education reduces mortality, but only for individuals whose unobserved characteristics that make them relatively less likely to attend college. Thus there is evidence of essential heterogeneity in the treatment of college on mortality; men with the largest potential gains were those who did not attend college.

My findings, in conjunction with prior research on the impact of college education on income, suggest that income is not the channel through which education reduces mortality. Longevity benefits are highest for the individuals that have the smallest income gains. As Deaton [2013] argues, it may be that "education directly promotes health ... because the knowledge and life lessons learned in school and college enable people to take better care of themselves and to take good advantage of the health care system when they need it." For example, a recent study by Case and Deaton [2015] shows that mortality rates for whites aged 45–54 has been increasing over the last 15 years, in part due to increase in opioid use. They find this mortality increase has been confined to individuals with high school education or less.

Cutler et al. [2006] hypothesize that the "greater speed of introduction of new health-relevant knowledge and technology will tend to raise the health gradient ..." If so, my results may well generalize to current cohorts of young people. My study of MTEs indicates that policies that increase college access at the margin will have health benefits for the men induced into higher education. For women, my results are less optimistic in terms of predicting improvements in women's longevity, but college attendance may nonetheless have important payoffs in terms of the health of the next generation.[27]

On the other hand, generalization on the basis of historical precedent is necessarily speculative. If these causal effects are driven primarily by the role of education in shifting men out of dangerous factory and mining jobs, college may now have a smaller effect on health outcomes than it did for previous generations, as these jobs become less common (and also likely safer). Changes in health care and medicine in the next 40 and 50 years could have dramatic and unpredictable impacts on any possible old-age mortality effects for current youth. Any efforts to predict how these changes will affect the mortality gains from education are fraught. Further research

[27] Currie and Moretti [2003] show that maternal education at the college level increases infant health.

on the mechanisms driving the results found in this paper is crucial for evaluating the external validity for contemporary generations.

References

Daniel Aaronson, Bhash Mazumder, Seth G. Sanders, and Evan J. Taylor. Estimating the effect of school quality on mortality in the presence of migration: Evidence from the Jim Crow south. *Working Paper*, 2016.

Valerie Albouy and Laurent Lequien. Does compulsory education lower mortality? *Journal of Health Economics*, 28(1):155–168, 2009.

Douglas Almond and Janet Currie. Human capital development before age 5. In David Card and Orley Ashenfelter, editors, *Handbook Labor Economics*, pages 1315–1486. Elsevier, Amsterdam, 2011.

Joshua D. Angrist, Guido W. Imbens, and Donald B. Rubin. Identification of causal effects using instrumental variables. *Journal of the American Statistical Association*, 91(434):444–455, 1996.

David J.P. Barker. *Mothers, Babies and Health in Later Life, 2nd ed.* Churchill Livingston, Edinburgh, 1998.

Gary Becker. *Human Capital and the Personal Distribution of Income*. University of Michigan Press, Ann Arbor Michigan, 1967.

Dan A. Black, Yu-Chieh Hsu, and Lowell J. Taylor. The effect of early-life education on later-life mortality. *Journal of Health Economics*, 44:1–9, 2015a.

Dan A. Black, Seth G. Sanders, Evan J. Taylor, and Lowell J. Taylor. The impact of the great migration on mortality of African Americans: Evidence from the deep south. *American Economic Review*, 105(2):477–503, 2015b.

Dan A. Black, Joonhwi Joo, Robert LaLonde, Jeffrey Smith, and Evan J. Taylor. Simple tests for selection bias: Learning more from instrumental variables. *IZA Discussion Paper, No. 9346*, 2016.

Christian N. Brinch, Magne Mogstad, and Matthew Wiswall. Beyond late with a discrete instrument: Heterogeneity in the quantity-quality interaction of children. *Journal of Political Economy*, forthcoming, 2016.

Kasey Buckles, Andreas Hagemann, Ofer Malamud, Melinda Morrill, and Abigail Wozniak. The effect of college education on mortality. *Journal of Health Economics*, 50:99–114, 2016.

Stephen V. Cameron and Christopher Taber. Estimation of education borrowing constraints using returns to schooling. *Journal of Political Economy*, 112(1):132–182, 2004.

David Card. Earnings, schooling and ability revisited. *NBER Working Paper No. 4832*, 1994.

David Card. Using geographic variation in college proximity to estimate the return to schooling. *Aspects of Labour Market Behaviour : Essays in Honour of John Vanderkamp*, pages 201–222, 1995.

David Card. The causal effect of education on earnings. In Orley Ashenfelter, Richard Layard, and David Card, editors, *Handbook of Labor Economics, vol 3*. Elservier, Amsterdam, 1999.

David Card. Estimating the return to schooling: Progress on some persistent econometric problems. *Econometrica*, 69(5):1127–1160, 2001.

Pedro Carneiro, James J. Heckman, and Edward J. Vytlacil. Estimating the marginal returns to education. *American Economic Review*, 101(6):2754–2781, 2011.

Anne Case and Angus Deaton. Rising morbidity and mortality in midlife among white non-hispanic Americans in the 21st century. *Proceedings of the National Academy of Science*, 112(49):15078–15083, 2015.

Damon Clark and Heather Royer. The effect of education on adult mortality and heatlh: Evidence from Britain. *American Economic Review*, 103(6):2087–2120, 2013.

Gabriella Conti, James J. Heckman, and Sergio Urzua. The education-health gradient. *American Economics Review: Papers and Proceedings*, 2010(2):234–238, 2010.

Janet Currie. Healthy, wealthy, and wise? Socioeconomic status, poor health in childhood, and human capital development. *Journal of Economic Literature*, 47(1): 87–122, 2009.

Janet Currie. Inequality at birth: Some causes and consequences. *American Economic Review*, 101(3):1–22, 2011.

Janet Currie and Enrico Moretti. Mother's education and the intergenerational transmission of human capital: Evidence from college openings. *The Quarterly Journal of Economics*, 118(4):1495–1532, 2003.

David Cutler, Angus Deaton, and Adriana Lleras-Muney. The determinants of mortality. *Journal of Economic Perspectives*, 20(3):97–120, 2006.

David M. Cutler and Adriana Lleras-Muney. Understand differences in health behaviors by education. *Journal of Health Economics*, 29(1):1–28, 2010.

Angus Deaton. What does the empirical evidence tell us about the injustice of health inequalities. In Nir Eyal, Samia Hurst, Ole Frithof Norheim, and Daniel Wikler, editors, *Inequalities in Health: Concepts, Measures and Ethics*, pages 263–281. Oxford University Press, Oxford, 2013.

Robert W. Fogel. New findings on secular trends in nutrition and mortality: Some implications for population theory. In Mark Rosenzweig and Oded Stark, editors, *Handbook of Population and Family Economics*, volume 1A, pages 433–481. Elsevier, Amsterdam, 1997.

Robert W. Fogel. *The Escape from Hunger and Premature Death, 1700-2100: Europe, America and the Third World*. Cambridge University Press, Cambridge, 2004.

Zvi Griliches. Estimating the returns to schooling: Some econometric problems. *Econometrica*, 45(1):1–22, 1977.

Franque Grimard and Daniel Parent. Education and smoking: Were Vietnam war draft avoiders also more likely to avoid smoking? *Journal of Health Economics*, 26 (5):896–926, 2007.

Michael Grossman. On the concept of health capital and the demand for health. *Journal of Political Economy*, 80(2):223–255, 1972.

Michael Grossman. The human capital model. In Anthony J. Culyer and Joseph P. Newhouse, editors, *Handbook of Health Economics*, volume 1, pages 347–408. Elsevier, Amsterdam, 2000.

James J. Heckman and Edward Vytlacil. Local instrumental variables and latent variable models for identifying and bounding treatment effects. *Proceedings of the National Academy of Sciences*, 96(8):4730–4734, 1999.

James J. Heckman and Edward Vytlacil. Structural equations, treatment effects, and econometric policy evaluation. *Econometrica*, 73(3):669–738, 2005.

James J. Heckman and Edward Vytlacil. Econometric evaluation of social programs, part ii. In James J. Heckman and Edward E. Leamer, editors, *Handbook of Econometrics*, volume 6B, pages 4875–5143. Elsevier, 2007.

James J. Heckman, Lance Lochner, and Petra E. Todd. Earnings functions, rates of return and treatment effects: The Mincer equation and beyond. In *Handbook of the Economics of Education*, volume 1, pages 307–458. Elsevier, Amsterdam, 2006a.

James J. Heckman, Sergio Urzua, and Edward Vytlacil. Understanding instrumental variables in models with essential heterogeneity. *The Review of Economics and Statistics*, 88(3):389–432, 2006b.

James J. Heckman, John Eric Humphries, and Gregory Veramendi. Dynamic treatment effects. *Journal of Econometrics*, 191:276–292, 2016.

Janna Johnson and Evan J. Taylor. The long-run consequences of urban-rural migration. *Working Paper*, 2016.

Adreana Lleras-Muney. The relationship between education and adult mortality in the United States. *The Review of Economic Studies*, 72(1):189–221, 2005.

Lance Lochner. Non-production benefits of education: Crime, health, and good citizenship. In Eric A. Hanushek, Stephen J. Machin, and Ludger Woessmann, editors, *Handbook of the Economics of Education*, volume 4. North Holland, Amsterdam, 2011.

Ofer Malamud and Abigail Wozniak. The impact of college education on geographic mobility. *Journal of Human Resources*, 47(4):913–950, 2012.

Bhashkar Mazumder. Does education improve health? A reexamination of evidence from compulsory schooling laws. *Economic Perspectives*, 32(2):2–16, 2008.

Ellen R. Meara, Seth Richards, and David M. Cutler. The gap gets bigger: Changes in mortality and life expectancy, by education, 1981-2000. *Health Affairs*, 27(2): 350–360, 2008.

Costas Meghir, Marten Palme, and Emilia Simeonova. Education, health, and mortality: Evidence from a social experiment. *NBER Working Paper No. 17932*, 2012.

Jacob Mincer. *Schooling, Experience, and Earnings*. National Bureau of Economic Research, New York, 1974.

Robert Moffitt. Estimating marginal treatment effects in heterogeneous populations. *Annales d'Economie et de Statistique*, (91-92):239–261, 2008.

Rati Ram and Theodore W. Schultz. Life span, health, savings, and productivity. *Economic Development and Cultural Change*, 27(3):399–421, 1979.

Mark R. Rosenzweig and T. Paul Schultz. The behavior of mothers as inputs to child health: The determinants of birth weight, gestation, and rate of fetal growth. In Victor R. Fuchs, editor, *Economic Aspects of Health*. University of Chicago Press, NBER, Chicago, 1982.

Arthur Roy. Some thoughts on the distribution of earnings. *Oxford Economic Papers*, 3(2):135–146, 1951.

Steven Ruggles, Katie Genadek, Ronald Goeken, Josiah Grover, and Matthew Sobek. *Integrated Public Use Microdata Series: Version 6.0 [Machine-readable database]*. University of Minnesota, Minneapolis, 2015.

James P. Smith. Healthy bodies and thick wallets: The dual relation between health and economic status. *Journal of Economic Perspectives*, 13(2):145–166, 1999.

Elpidoforos S. Soteriades and Joseph R. DiFranza. Parent's socioeconomic status, adolescents' disposable income, and adolescents' smoking status in massachusetts. *American Journal of Public Health*, 93(7):1155–1160, 2003.

James H. Stock, Jonathan W. Wright, and Motohiro Yogo. A survey of weak in-

struments and weak identification in generalized method of moments. *Journal of Business and Economic Statistics*, 17(3):518–529, 2002.

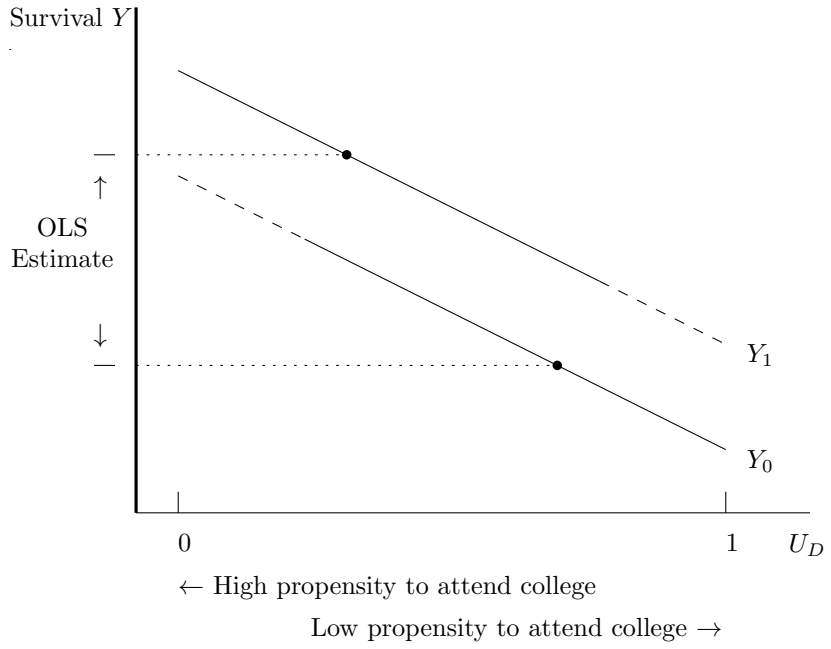
Evan J. Taylor. Comparing estimation of marginal treatment effects with local instrumental variables and the separate approach. *working paper*, 2016.

Edward Vytlacil. Independence, monotonicity, and latent index models: An equivalence result. *Econometrica*, 70(1):331–341, 2002.

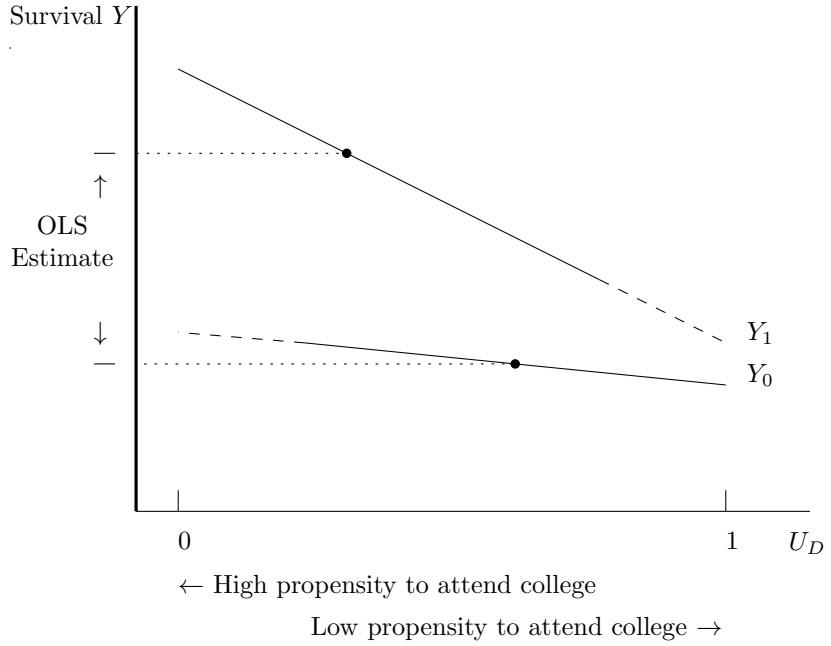
Damien De Walque. Does education affect smoking behaviors? Evidence using the Vietnam draft as an instrument for college education. *Journal of Health Economics*, 26(5):877–895, 2007.

Figure 1: Patterns of Selection into Treatment (College Attendance)

A. Ability Bias



B. Selection on Gains from Treatment



C. Treatment Effect Highest among those Selecting Out of Treatment

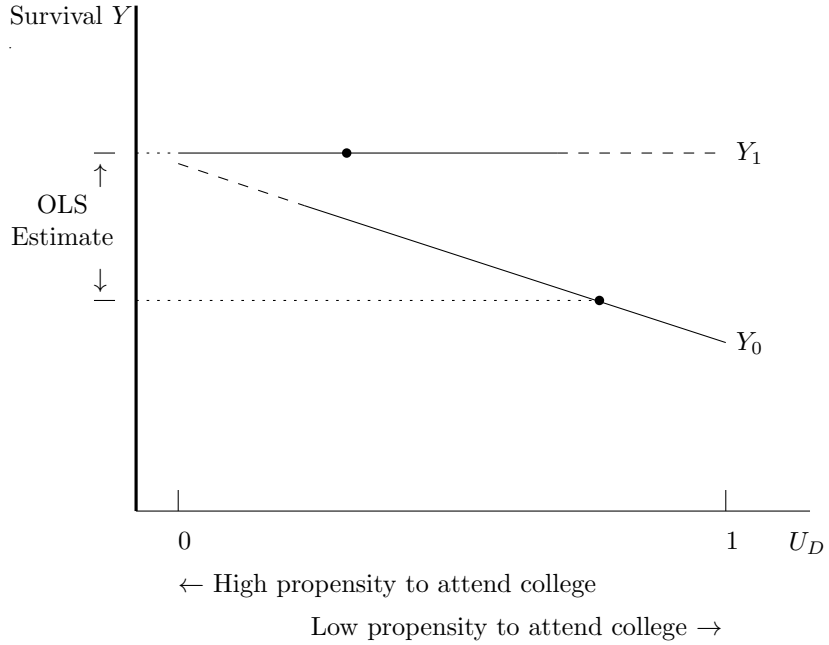
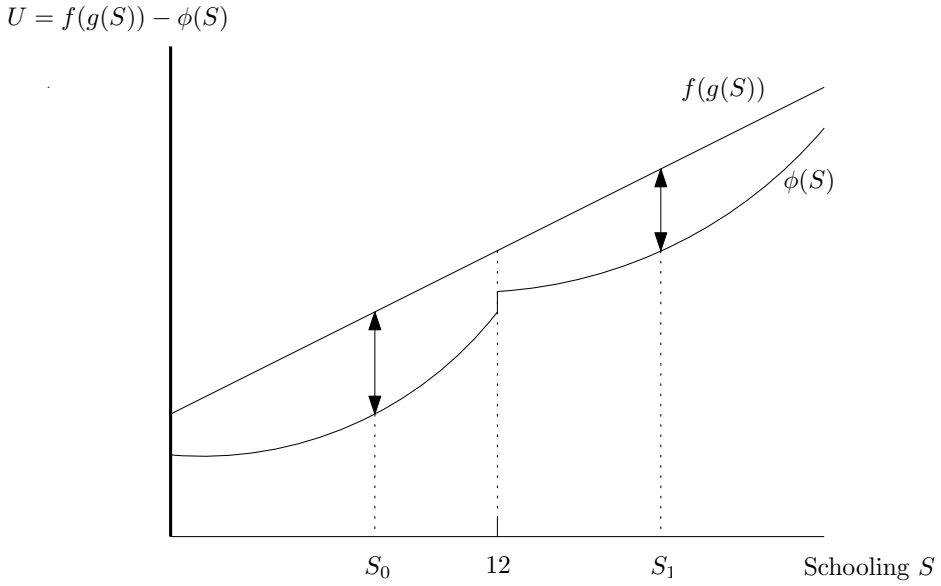
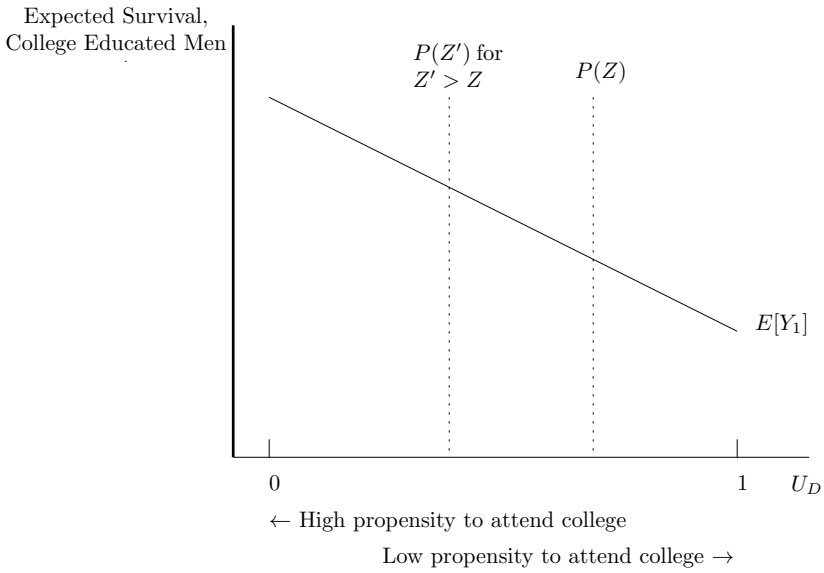


Figure 2: An Example of Benefit and Cost Curves for Schooling Choice



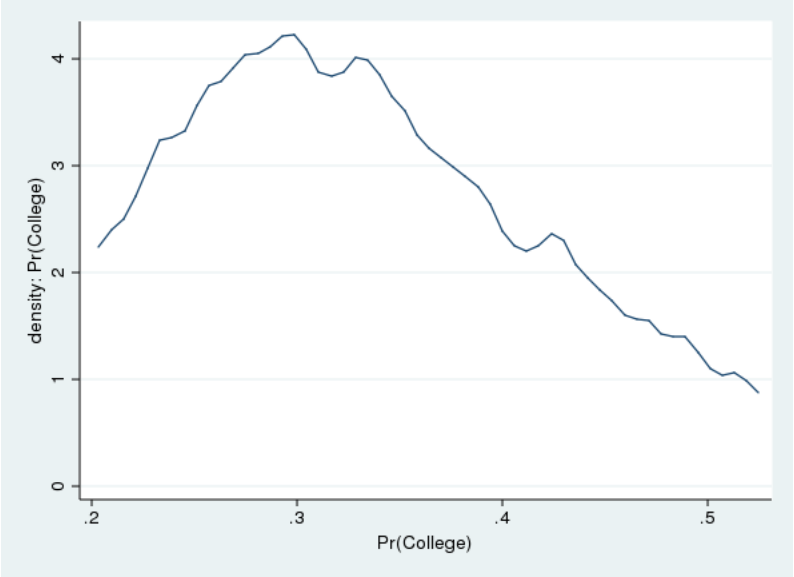
The instrument can only induce parallel shifts of the cost curve, $\phi(S)$ for $S > 12$, and have no effect on the benefit curve, $f(g(S))$. S_0 shows the optimal choice of schooling for $S \leq 12$ and S_1 shows the optimal choice of schooling for $S > 12$.

Figure 3: An Example of Positive Selection on the Treated Outcome



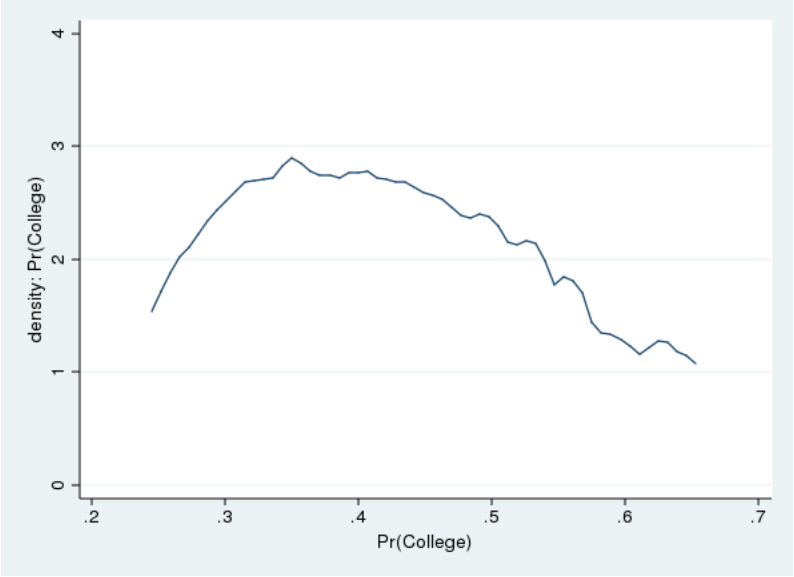
When P = P(Z), all individuals to left of the right most vertical line become treated. When P = P(Z') the indifference line shifts left, which increases the average outcome for treated individuals.

Figure 4: Histogram of Propensity Scores for College, White Women born 1911-1940



Density is trimmed at 5 pct on both sides per Census guidelines

Figure 5: Histogram of Propensity Scores for College, White Men born 1911-1940



Density is trimmed at 5 pct on both sides per Census guidelines

Table 1: Correlations between County Level Covariates

	Median Housing Value, 1930	Radio Percent, 1940	Median Family Income, 1950
Median Housing Value, 1930	1		
Percent of HH with Radio, 1940	0.585	1	
Median Family Income, 1950	0.575	0.832	1

Correlation coefficients between county level covariates, not weighted by population. Median Family Income is measured in deciles, with ten being the highest and one being the lowest

Table 2: The Black et al. [2016] Selection Test

	Women		Men	
	Treated (College)	Untreated (HS or Less)	Treated (College)	Untreated (HS or Less)
College Proximity (100s Miles)	0.00582	0.00681	-0.00278	0.00906
	(0.00359)	(0.00376)	(0.00357)	(0.00381)
Two-sided P-Value	0.105	0.070	0.436	0.018
Observations	923,000	1,786,000	954,000	1,266,000

Estimated OLS regression coefficients on the instrument. The dependent variable is ten-year survival. Regressions are estimated separately for those with at least some college ("College") and those with high school education or less. County level economic covariates, birth state and cohort fixed effects are included.

Table 3: OLS Results on 10-year Survival, Women

	(1)	(2)	(3)	(4)
College Attendance	0.0625	0.0607	0.0613	0.0609
	(0.0007)	(0.0007)	(0.0007)	(0.0007)
Avg Per Capita Income (1969–2010)				0.0002
				(0.0002)
Median Housing Value (1930)				0.0006
				(0.0004)
Percent of HHs with Radio (1940)				0.0030
				(0.0081)
Cohort FE	x	x	x	x
Birth State FE		x		x
Birth County FE			x	
Observations (rounded)	2,709,000	2,709,000	2,709,000	2,709,000
R-squared	0.186	0.188	0.189	0.188
Clusters	3103	3103	3103	3103

Standard errors are clustered at the birth-county level. Mortality is measured from 2000–2010. "College Attendance" indicates at least some college. Fixed effects (FE) in deciles of 1950 Median Family Income also included in (4). Mean of the dependent variable is 0.69.

Table 4: OLS Results on 10-year Survival, Men

	(1)	(2)	(3)	(4)
College Attendance	0.0855	0.0833	0.0841	0.0833
	(0.0011)	(0.0009)	(0.0009)	(0.0009)
Avg Per Capita Income (1969–2010)				0.0002
				(0.0002)
Median Housing Value (1930)				0.0006
				(0.0005)
Percent of HHs with Radio (1940)				-0.0185
				(0.0081)
Cohort FE	x	x	x	x
Birth State FE		x		x
Birth County FE			x	
Observations (rounded)	2,220,000	2,220,000	2,220,000	2,220,000
R-squared	0.180	0.185	0.187	0.185
Clusters	3105	3105	3105	3105

Standard errors are clustered at the birth-county level. Mortality is measured from 2000–2010. "College Attendance" indicates at least some college. Fixed effects (FE) in deciles of 1950 Median Family Income also included in (4). Mean of the dependent variable is 0.62.

Table 5: Linear First Stage on College, Women

	(1)	(2)	(3)
College Proximity (100s Miles)	-0.0296	-0.0947	-0.1682
	(0.0059)	(0.0130)	(0.0225)
Proximity Squared		0.0814	0.2800
		(0.0130)	(0.0460)
Proximity Cubed			-0.1190
			(0.0244)
Avg Per Capita Income (1969-2010)	-0.0005	-0.0004	-0.0003
	(0.0005)	(0.0005)	(0.0004)
Median Housing Value (1930)	0.0107	0.0100	0.0095
	(0.0011)	(0.0011)	(0.0011)
Percent of HHs with Radio (1940)	0.1454	0.1427	0.1451
	(0.0206)	(0.0204)	(0.0203)
Observations (rounded)	2,709,000	2,709,000	2,709,000
Partial F-Statistic	25.2	26.7	21.7
Clusters	3103	3103	3103

All regression include birth-state and cohort fixed effects, and deciles of birth-county 1950 Median Family Income. The dependent variable is an indicator equal to one for at least some college. College proximity measures the distance from the birth-county to the nearest college. Standard errors are clustered at the birth-county level.

Table 6: Linear First Stage on College, Men

	(1)	(2)	(3)
College Proximity (100s Miles)	-0.0506	-0.1390	-0.2438
	(0.0083)	(0.0179)	(0.0325)
Proximity Squared		0.1110	0.3940
		(0.0172)	(0.0650)
Proximity Cubed			-0.1700
			(0.0329)
Avg Per Capita Income, (1969-2010)	-0.0033	-0.0032	-0.0031
	(0.0010)	(0.0010)	(0.0009)
Median Housing Value (1930)	0.0229	0.0220	0.0213
	(0.0020)	(0.0019)	(0.0019)
Percent of HHs with Radio (1940)	0.2557	0.2522	0.2552
	(0.0296)	(0.0293)	(0.0292)
Observations (rounded)	2,220,000	2,220,000	2,220,000
Partial F-Statistic	36.9	30.7	22.4
Clusters	3105	3105	3105

All regression include birth-state and cohort fixed effects, and deciles of birth-county 1950 Median Family Income. The dependent variable is an indicator equal to one for at least some college. College proximity measures the distance from the birth-county to the nearest college. Standard errors are clustered at the birth-county level.

Table 7: IV Results on 10-year Survival

	Women			Men		
	(1)	(2)	(3)	(4)	(5)	(6)
Deg(Z), 1st Stage:	Linear	Quadratic	Cubic	Linear	Quadratic	Cubic
College Attendance	-0.1561	-0.0509	-0.0558	-0.0289	0.0321	0.0537
	(0.1059)	(0.0670)	(0.0614)	(0.0567)	(0.0423)	(0.0404)
Avg Per Capita Inc (1970-2010)	0.0000	0.0001	0.0001	-0.0002	0.0000	0.0001
	(0.0002)	(0.0002)	(0.0002)	(0.0004)	(0.0003)	(0.0003)
Median Housing Value (1930)	0.0030	0.0019	0.0020	0.0033	0.0019	0.0014
	(0.0013)	(0.0009)	(0.0008)	(0.0015)	(0.0011)	(0.0011)
Pct of HHs with Radio (1940)	0.0377	0.0209	0.0217	0.0130	-0.0042	-0.0102
	(0.0194)	(0.0136)	(0.0129)	(0.0179)	(0.0142)	(0.0138)
Obs. (rounded)	2,709,000	2,709,000	2,709,000	2,220,000	2,220,000	2,220,000
Clusters	3103	3103	3103	3105	3105	3105

All regressions include birth-state and cohort fixed effects, and deciles of birth-county 1950 Median Family Income. Mortality is measured from 2000–2010. College Attendance indicates at least some college. Standard errors are clustered at the birth-county level.

Table 8: Estimated Slope of the MTE of College Attendance on Survival

Panel A. Women

First Stage:	(1) Flexible Logit	(2) Linear Probability	(3) Logit
$\hat{\beta}$	0.052	0.059	0.082
	(0.040)	(0.046)	(0.046)

Panel B. Men

First Stage:	(1) Flexible Logit	(2) Linear Probability	(3) Logit
$\hat{\beta}$	0.086	0.074	0.079
	(0.050)	(0.062)	(0.061)

All regressions include birth-state and cohort fixed effects, and deciles of birth-county 1950 Median Family Income. Mortality measured from 2000–2010. College Attendance indicates at least some college. Standard errors are clustered at the birth-county level.

Table 9: Parameter Estimates of $\Delta^{MTE}(X, U_D) = \alpha + \beta U_D + \gamma X$, with and without Heterogeneity in Observables X

Panel A. Women

	(1) Constant Effect	(2) Local IV	(3) Local IV	(4) Separate Approach	(5) Separate Approach
$\hat{\alpha}$	-0.003	0.086	-0.094	0.014	-0.042
	(0.024)	(0.048)	(0.118)	(0.025)	(0.030)
$\hat{\beta}$		-0.241	0.597	-0.049	0.052
		(0.102)	(0.396)	(0.014)	(0.040)
$\gamma = 0$	yes	yes	no	yes	no

Panel A. Men

	(1) Constant Effect	(2) Local IV	(3) Local IV	(4) Separate Approach	(5) Separate Approach
$\hat{\alpha}$	0.028	0.004	-0.067	-0.012	0.005
	(0.018)	(0.038)	(0.106)	(0.019)	(0.027)
$\hat{\beta}$		0.061	0.291	0.085	0.086
		(0.080)	(0.285)	(0.014)	(0.050)
$\gamma = 0$	yes	yes	no	yes	no

Mortality measured from 2000–2010. Standard errors are clustered at the birth-county level.

Table 10: Average Treatment Effects for the Treated and Untreated

Panel A. Women

	Untreated	Treated	Difference
Average Y_0	0.665	0.739	0.074
	(0.003)	(0.031)	(0.030)
Average Y_1	0.679	0.728	0.050
	(0.028)	(0.003)	(0.029)
Average Treatment, $Y_1 - Y_0$	0.014	-0.010	-0.024
	(0.028)	(0.030)	(0.035)
Observations	1,786,000	923,000	

Panel B. Men

	Untreated	Treated	Difference
Average Y_0	0.586	0.666	0.080
	(0.003)	(0.025)	(0.024)
Average Y_1	0.648	0.668	0.020
	(0.023)	(0.003)	(0.023)
Average Treatment, $Y_1 - Y_0$	0.062	0.002	-0.060
	(0.023)	(0.025)	(0.032)
Observations	1,266,000	954,000	

Birth cohorts are weighted for balance across treatment and untreated. Calculations are conducted using extrapolation of the Marginal Treatment Effect. Standard errors clustered at the county level.

Table 11: Average Treatment Effects for the Treated and Untreated with Balanced Observables

Panel A. Women

	Untreated	Treated	Difference
Average Y_0	0.667	0.737	0.070
	(0.003)	(0.031)	(0.029)
Average Y_1	0.680	0.727	0.047
	(0.028)	(0.003)	(0.028)
Average Treatment, $Y_1 - Y_0$	0.013	-0.010	-0.023
	(0.028)	(0.030)	(0.034)
Observations	1,786,000	923,000	

Panel B. Men

	Untreated	Treated	Difference
Average Y_0	0.589	0.663	0.075
	(0.003)	(0.024)	(0.023)
Average Y_1	0.650	0.666	0.016
	(0.022)	(0.003)	(0.022)
Average Treatment, $Y_1 - Y_0$	0.061	0.003	-0.058
	(0.023)	(0.025)	(0.030)
Observations	1,266,000	954,000	

All observed covariates are balanced across the treated and untreated. Calculations are based on extrapolation of the Marginal Treatment Effect. Standard errors clustered at the county level.

Table 12: Gender and Education in Selected Dangerous Occupations, 1960

	Men		Women	
Occupation	College	HS or Less	College	HS or Less
Furnacemen, smeltermen, and pourers	75	1,375	1	41
Mine operatives and laborers, n.e.c.	595	10,481	5	48
Motormen, mine, factory, logging camp, etc.	6	415	0	1
Total observations	328,869	915,512	254,468	1,029,656

Author's calculations using the 1960 IPUMS, Ruggles et al. [2015]. Counts are for whites born in the U.S., 1911–1940. "College" indicates at least some college.

Table 13: OLS Regressions of Location and Survival Rates

Panel A. Women

		Dependent Variable	
	Live Outside Birth State	10-Year Survival	10-Year Survival
College Attendance	0.1464		0.0623
	(0.0037)		(0.0007)
Live Outside Birth State		-0.0008	-0.0095
		(0.0009)	(0.0008)
Observations (rounded)	2,709,000	2,709,000	2,709,000
Clusters	3103	3103	3103

Panel B. Men

		Dependent Variable	
	Live Outside Birth State	10-Year Survival	10-Year Survival
College Attendance	0.1912		0.0844
	(0.0038)		(0.0009)
Live Outside Birth State		0.0106	-0.0058
		(0.0008)	(0.0007)
Observations (rounded)	2,220,000	2,220,000	2,220,000
Clusters	3105	3105	3105

"Live Outside Birth State" is an indicator equal to 1 if an individual lives in a different state than the state of birth as of 2000. Mortality measured from 2000–2010. All regression include birth-state and cohort fixed effects. Standard errors clustered at birth-county level.